

SECTION 33 12 73
HORIZONTAL CENTRIFUGAL PUMPING UNIT

PART 1 GENERAL

1.01 REFERENCE STANDARDS

- A. American Bearing Manufacturers Association (ABMA)
1. ABMA 9-2015 (R2020) Load Ratings and Fatigue Life for Ball Bearings
 2. ABMA 11-2014 (R2020) Load Ratings and Fatigue Life for Roller Bearings
- B. American Society of Civil Engineers (ASCE)
1. ASCE 7-2022 Minimum Design Loads for Buildings and Other Structures
- C. American Society of Mechanical Engineers (ASME)
1. ASME BPVC-2025 Boiler and Pressure Vessel Code
 2. ASME B16.5-2025 Pipe Flanges and Flanged Fittings –
NPS ½ Through NPS 24 Metric/Inch
Standard
- D. ASTM International (ASTM)
1. ASTM A 108/A 108M-2024 Steel Bar, Carbon and Alloy, Cold-Finished
 2. ASTM A 536-2024 Ductile Iron Castings
 3. ASTM A 48/A 48M-2022 Gray Iron Castings
 4. ASTM B 148-2024 Aluminum-Bronze Sand Castings
 5. ASTM B 505/B 505M-2023 Copper Alloy Continuous Castings
- E. American Water Works Association (AWWA)
1. AWWA C207-2023 Steel Pipe Flanges for Waterworks Service, Sizes 4 in. Through 144 in. (100 mm Through 3,600 mm)
- F. American Welding Society (AWS)
1. AWS D1.1/D1.1M-2025 Structural Welding Code - Steel
- G. Hydraulic Institute Standards (HI)

1. HI 9.1-9.5-2021 Pumps General Guidelines –
for Materials, Sound Testing, and
Decontamination
2. HI 9.6.1-2024 Rotodynamic Pumps -
Guideline for NPSH Margin
3. HI 9.6.3-2024 Rotodynamic Pumps
– Guideline for Operating Regions
4. HI 9.6.4-2022 Rotodynamic Pumps
– for Vibration Measurements and
Allowable Values
5. HI 9.8-2024 Rotodynamic Pumps -
for Pump Intake Design
6. HI 14.1-14.2-2024 Rotodynamic Pumps - for
Nomenclature and Definitions
7. HI 14.3-2024 Rotodynamic Pumps -
for Design and Application
8. HI 14.4-2024 Rotodynamic Pumps -
for Installation, Operation, and Maintenance
9. HI 14.6-2022 Rotodynamic Pumps -
for Hydraulic Performance Acceptance
Tests
- H. International Electrotechnical Commission (IEC)
 1. IEC 61800 Adjustable Speed
Electrical Power Drive Systems
- I. Institute of Electrical And Electronics Engineers (IEEE)
 1. IEEE 43-2013
Recommended Practice for Testing
Insulation Resistance of Rotating Machinery
 2. IEEE 112-2017 Test
Procedure for Polyphase Induction Motors
and Generators
 3. IEEE 519-2022
Harmonic Control in Electric Power
Systems
- J. International Standard Organization (ISO)
 1. ISO 1940/1-2003 Mechanical Vibration
- K. National Electrical Manufacturers Association (NEMA)

1. NEMA ICS 7-2020 Industrial Control and Systems: Adjustable Speed Drives
 2. NEMA KS 1-2013 Heavy Enclosed and Dead-Front Switches (600 Volts Maximum)
 3. NEMA MG 1-2024 Motors and Generators (Includes Errata 2012)
 4. NEMA IS 10033-2020 (R2025) Adjustable Speed Drives
- L. Underwriters Lab (UL)
1. UL 508-2024 Equipment UL Standard for Safety Industrial Control
- M. National Fire Protection Association (NFPA)
1. NFPA 70-2023 National Electric Code

1.02 DEFINITIONS

- A. Brinell hardness (BHN): A measure of the resistance to indentation of a material, as defined in ASTM E10.
- B. NPSHa: Net Positive Suction Head Available.
- C. NPSHr: Net Positive Suction Head Required.
- D. Pumping Unit: Pump, motor, and baseplate.
- E. RTD: Resistance Temperature Device.
- F. TDH: Total Dynamic Head.
- G. VFD: Variable Frequency Drive.

1.03 SYSTEM DESCRIPTION

- A. Background:
1. The low ambient pump unit background is provided in the project Statement of Work (SOW).
- B. Design Requirements:
1. Horizontal-shaft, split-case, single-stage, double-suction centrifugal type pump, complete with accessories, piping, anchor bolt assemblies, tools, and spare parts, suitable for direct connection through a flexible coupling to a horizontal induction motor.
 2. Design and construct pump in accordance with these specifications and HI 14.1-14.4, 9.1-9.5, 9.6.1-9.6.4, 9.8, and 14.3.

3. Mount pump and motor on a common baseplate supplied by the pump manufacturer.
4. Weight of pumping unit rotating parts, including unbalanced hydraulic thrust of impeller, carried by the pump bearing.
5. Direction of rotation of the pumping unit, when facing the motor end of the pump, to be as listed in Table 33 12 73A – Performance Requirements.
6. Pumping unit to be suitable for indoor operation at elevation of 5,650-feet and with an ambient temperature of 104-degrees Fahrenheit.
7. Pump inlet and discharge flanges to be compatible with piping flanges and ASME B16.5.
8. Pump flange-to-flange distance not to exceed 54-inch.
9. Normal operating conditions:
 - a. Starting of the unit will be controlled from pump’s individual VFD.
10. Pumping unit shall be seismically restrained to meet ASCE 7 Chapter 13 – *Seismic Design Requirements for Nonstructural Components*. Calculations should take into account all forces and loads including seismic that the pumping unit will be subjected to.

C. Performance Requirements:

1. Refer to Table 33 12 73A - Performance Requirements.
2. Operation of the pump at rated total head, shown in Table 33 12 73A – Performance Requirements, to be within the preferred operating range (POR).
3. The efficiency of the pump when operating at rated total head shown in Table 33 12 73A - Performance Requirements not to be less than 80-percent for operating conditions.
4. Select pump with a head-capacity curve that has a continuously decreasing head with increasing capacity over the allowable operating range (AOR).
5. The pump minimum net positive suction head available (NPSHa) of approximately 33 feet shall exceed the pump net positive suction head required (NPSHr) for the pump by the amount of margin and margin ratio specified in HI 9.6.1.

Table 33 12 73A - Performance Requirements

Direction of Rotation*	Motor Capacity (HP)	Unit rated capacity at rated total head (gpm)	Rated total head (feet)
CW	100	4,500	60

* Viewed from driver side.

1.04 DELIVERABLE REQUIREMENTS

- A. The following are requirements for the applicable deliverables listed in the Statement of Work (SOW).
- B. Datasheets:
 - 1. Pump
 - a. Include commercial product data including manufacturer's name, type, model, size, dimensions, weights, and materials of construction for each component.
 - b. Include total dynamic head versus flow curves, with head in feet and flow in gallons per minute (gpm). Power, efficiency, and net positive suction head required (NPSHr) are to be included on the same scale of flow, with power in horsepower and NPSHr in feet.
 - 1) Include multiple curves at varying rotational speeds, in rpm.
 - 2) Indicate POR and AOR on the pump curves, per HI 9.6.3.
 - c. Include impeller diameter, speed, and percentage of best efficiency point (BEP) at the rated point.
 - d. Include specific speed (Ns) and suction specific speed (Nss).
 - 2. Motor
 - a. Catalog data, including complete identifying data giving manufacturer's name, type, model, size and dimensions for:
 - 1) Electrical terminal boxes.
 - 2) Motor bearing temperature sensors and relays.
 - 3) Winding temperature elements/sensors.
 - b. Assembly and sectional drawings with parts and materials lists, and component weights.
 - c. Motor outline drawings including thrust bearings and dimensioned location of terminal boxes.
 - d. Manufacturer's recommended tightening torques for the motor mounting bolts.
 - e. Electrical, schematic, and wiring diagrams for motor and controls.
 - f. Description of motor insulation.
 - g. Idle (motor stopped) time required to cool from rated load temperature to ambient temperature. Running time, fully loaded, after two consecutive cold starts of motor before a third start may be initiated.
 - h. Nameplate data.
 - i. Factory motor test reports.

3. Variable Frequency Drive
 - a. Manufacturer's Technical Data.
 - b. Schematic diagram, wiring diagram, nameplate list, and bill of materials.
 - c. Factory test report.
4. Fused Disconnect Switch
 - a. Manufacturer's technical data.
5. Cable
 - a. Manufacturer's technical data.
- C. Installation Plan:
 1. Install pump and motor on a common baseplate.
 - a. In accordance with Drawing 40-D-6006.
 2. Include manufacturer's recommended tightening torques for the baseplate anchor bolts and pump and motor mounting bolts.
 3. Include leveling and grouting of the baseplate to the foundation.
 4. Include detailed installation and alignment procedures, per HI 14.4.
 5. Include detailed schematic and wiring diagrams for Fused Disconnect Switch, VFD, Motor and Emergency Stop Switch.
- D. Field Test Plan:
 1. Provide a detailed pumping unit field test plan.
- E. Operations and Maintenance Manual:
 1. Operation and maintenance instructions, per HI 14.4, including tools and accessories for assembly and disassembly of the pump and associated components.
 - a. Include a maintenance schedule in tabular form.
- F. Field Test Report:
 1. Provide a report of tests which shall include:
 - a. Drawing showing specific locations of vibration probes and bearing temperature sensors.
 - b. Vibration analysis, noise level, shaft alignment and run-out measurements, pump and motor bearing temperatures, motor voltage and amperage, and motor stator temperature.
 - c. Presentation of vibration data in a summary form and as a comparison with specification requirements for allowable vibration limit.
 - d. Observations and equipment adjustments performed during testing.

- e. Detailed discussion and summary of conclusions drawn from vibration, motor bearing and winding temperature, and motor voltage and amperage data that was recorded.
- f. Detailed summary of test results of VFD operation and SCADA control of VFD remote operation.

1.05 DELIVERY, STORAGE, AND HANDLING

- A. Prevent damage to pumping unit components and associated equipment during loading, transportation, unloading, and storage.
- B. Repair damage to the pumping unit and associated equipment as directed if, in the opinion of the Government, a satisfactory repair can be made; otherwise replace damaged pumping unit components and associated equipment.
- C. Protect from corrosion, deformation and other types of damage.

1.06 QUALIFICATIONS

- A. Pump Manufacturer Qualifications:
 - 1. Provide documentation of similar pumping unit designed and manufactured with five (5) years of successful operation.
 - 2. Provide the following information:
 - a. Manufacturing location city and state.
 - b. Customer names, addresses, telephone numbers, and names of projects.
 - c. Application type for each pumping unit.
 - d. Pump capacity, total head, rotational speed, impeller size, and working fluid.
- B. Welding Qualifications:
 - 1. Welding procedures, welders, and welding operators:
 - a. ASME BPVC Section IX, or,
 - b. AWS D1.1, or,
 - c. equivalent.
- C. Pump and Motor Manufacturer Erection Engineer Qualifications:
 - 1. Pump and motor erection engineer can be one and the same.
 - 2. Minimum five (5) years of experience in the erection of pumping equipment, specifically pumps and motors, of similar size and type.
 - 3. Provide a project list of recent related projects that the erection engineer was directly involved in.

D. Millwright Qualifications:

1. Millwright qualifications to include a minimum five (5) years of experience installing rotating pumping equipment of similar scope and size.
2. Provide name, experience, and formal education of primary tradesman and assistants who will be performing the work.
3. Provide a project list of recent millwright related projects that the primary tradesman was directly involved in.
4. Provide a complete list of available precision levels and laser alignment tools used to perform work.

E. Electrician Qualifications:

1. Electrician qualifications to include a minimum of three (3) years of documented experience installing low voltage, as defined by NFPA 70, equipment of similar scope and size.
2. Journeyman electrician license with current completion of OSHA 10/30-hour training and NFPA 70 training.

F. Controls Technician Qualifications:

1. Controls technician qualifications to include minimum of five (5) years of documented experience with:
 - a. Crimson software, Red Lion PID Modules and protocols as well as Gateway/Converter knowledge.
 - b. VFD commissioning and motor setup to include drive start-up, communications configuration and integration of PLC-PID modules.

1.07 EXTRA MATERIALS

A. Spare Parts:

1. Provide spare parts that are “new” and not refurbished and properly packaged for long-term storage.
2. Furnish spare parts that are interchangeable with and of the same materials and workmanship as corresponding original pump and motor parts.
3. Clearly mark or tag each part to identify size and type of pumping unit it is intended. Specifically identify with part unit number and specific part number.
4. Furnish the one set of the following spare parts for each different size pumping unit:
 - a. Gaskets, seals, as applicable.
 - b. Packing.

B. Special Tools:

1. Furnish one set of special tools required for assembly and disassembly of the pumping unit (pump and motor) and other appurtenances and accessories that may be required for assembling, installing, aligning, or dismantling part of the pump and to make the unit complete and ready for operation. (Special tools are considered to be those tools, which because of their limited use, are not normally available but which are necessary for the particular equipment.)
2. Furnish tools of high-grade, forged alloy, tool steel.
3. Furnish major parts of the pump with eyebolts, lugs, or lifting devices (i.e., lifting beam) to facilitate handling with a mobile crane. Lifting equipment to comply with performance standards.

PART 2 PRODUCTS

2.01 EPOXY GROUT

- A. Three part flowable epoxy grout; resin, hardener, and aggregate filler:
1. CHOCKFAST RED manufactured by ITW Performance Polymers, 130 Commerce Drive, Montgomery PA; or equal, having the following essential characteristics:
 - a. ASTM C579 Compressive Strength: Minimum of 15,000 psi.
 - b. ASTM C579 Compressive Modulus: 2,000,000 psi.
 - c. ASTM D696 Coefficient of Linear Thermal Expansion:
 - 1) 11.2×10^{-6} degrees F.
 - d. ASTM C580 Flexural Strength: Minimum 4,000 psi.
 - e. ASTM D638 Tensile Strength: Minimum 1,800 psi.
 - f. Service temperature: Up to 140 degrees F.
 - g. Pot life: Approximately 3 hours at 70 degrees F.
 - h. Shelf life: 2 years in dry storage.
- B. Aggregate filler as recommended by manufacturer.

2.02 NAMEPLATES

- A. General:
1. Black lettering.
 2. Provide holes in each corner with screws for attachment to equipment.
- B. Motor Nameplate Requirements:
1. Provide new nameplate to include the following information, at a minimum:
 - a. Contractor Name and Logo

- b. Installed Year
- c. Model Number
- d. Serial Number
- e. Frame Size
- f. Horsepower
- g. Voltage, Amperage, Frequency (Hertz)
- h. Number of Poles
- i. Service Factor
- j. Number of Phases
- k. Speed (rotations per minute)

C. Pump Nameplate Requirements:

- 1. Provide new nameplate to include the following information, at a minimum:
 - a. Contractor Name and Logo
 - b. Installed Year
 - c. Model Number
 - d. Serial Number
 - e. Design TDH (feet)
 - f. Design Capacity (gpm)
 - g. Impeller Diameter (inches)
 - h. Speed (rpm)

D. VFD Nameplate Requirements:

- 1. Provide new nameplate to include the following information, at a minimum:
 - a. Contractor Name and Logo
 - b. Installed Year
 - c. Model Number
 - d. Serial Number
 - e. Input Voltage, Current, Frequency
 - f. Output Voltage, Current, Frequency
 - g. Rated Motor HP
 - h. Overload Capacity

2.03 HORIZONTAL CENTRIFUGAL PUMPING UNITS

A. Pump:

1. Casing:

- a. Provide cast iron or ductile iron casing construction.
- b. Double suction type.
- c. Design to produce smooth flow with gradual changes in velocity.
- d. Split on the horizontal centerline, with suction and discharge connections cast integral with the lower half.
- e. Suction and discharge connections to be faced and drilled to conform to AWWA C207.
- f. Design for easy removal of impeller and bearings.
- g. Provide with casing wearing ring of aluminum bronze material. Brinell hardness to be at least 100 BHN less than the impeller Brinell hardness.
- h. Provide suitable drilled and tapped holes for air release valve and drain connections.
 - 1) Provide a standard NPT tapped hole at top of casing for installation of an air release valve.
 - 2) Locate drains such that the casing can be completely unwatered.
- i. Casing construction to permit pump to be readily assembled and dismantled.
- j. Inspect castings thoroughly. Ensure that surfaces of vanes used to direct water flow through the suction case and pump bowl are as smooth as practicable and free of blow holes, chilled areas, slag, or foreign matter. Repair or reject defective castings.

2. Impeller:

- a. Enclosed type and made in one piece entirely of stainless-steel material with integral wear ring surfaces.
- b. Securely fastened to shaft in such a manner as to make it readily removable.
- c. Wearing rings shrink-fit to the impeller, utilizing an interference fit of 0.0005-inch per inch of ring inside diameter.
- d. Radial clearances between rotating and stationary wearing rings to be as small as possible consistent with safe operation and with the clearances required in the pump guide bearings.
- e. Hand-finish impeller water passages to remove rough spots and excessive irregularities.

- f. Statically and dynamically balance rotating parts to prevent whipping and vibration throughout operating range from shutoff head to run out.
 - g. Balance in accordance with ISO-1940/1-(E) to a balance quality grade of at least G6.3.
 - 1) Balance to pump manufacturer's quality control standards provided the residual imbalance during operational field testing performed after installation at the job site does not exceed 1/2 the vibration limit established by HI 9.6.4.
 - h. Inspect castings thoroughly. Ensure that surfaces of hub and blades are as smooth as practicable and free of blow holes, chilled areas, slag, or foreign matter. Repair or reject defective castings.
- 3. Shaft:
 - a. Provide stainless-steel shaft construction.
 - b. Provide a shaft of sufficient size to operate without objectionable distortion or vibration at maximum speed in both the forward and reverse directions.
 - 1) Size shaft to prevent torsional and flexural deflection which could cause whipping and vibration under condition.
 - 2) Shaft first critical speed to be at least 125-percent of operating speed.
 - c. Provide with removable and renewable shaft sleeves where it passes through the stuffing boxes and water passages.
 - 1) Gasketed to prevent water being pumped from contacting shaft.
 - 2) Secured in place for both directions of pump rotation.
 - 3) Made of stainless steel.
 - d. Provide water deflectors on shaft to prevent water from passing along shaft and entering pump bearings.
- 4. Packing Gland:
 - a. Suitable for the appropriate packing material and amount of packing rings necessary.
 - b. Appropriately sized for the shaft and any interfacing components.
 - c. Temperature rise, expected with extended operation, in the packing and shaft to not exceed an amount that would cause damage to components.
 - d. Design leakage rate to be between a steady drip and small stream.
- 5. Coupling and Guard:
 - a. Connect pump and motor shafts with a metal gear-type or disc-type, flexible coupling to compensate for parallel offset and angular misalignment.

- b. Flexible coupling to be the heavy-duty type sized for 1.5 times the motor horsepower nameplate rating.
 - c. Provide an all-metal OSHA-approved coupling guard that is removable and supported from the baseplate with stainless steel bolts and nuts.
- 6. Pump Bearings:
 - a. Support pump shaft by two bearings of suitable design with one located on each side of the impeller.
 - b. Removable in the field without damage to the bearings or shaft.
 - 1) Bearings requiring application of heat for removal not acceptable.
 - 2) Provide detailed procedure for removal of bearings in installation, operation, and maintenance instructions.
 - c. One bearing to be of the thrust type designed to carry the unbalanced hydraulic thrust the pump might develop.
 - d. Bearings to be grease-lubricated, ball- or roller-bearing type.
 - e. Connections for grease-lubricated bearings to be button-head type.
 - f. Bearings to not require external cooling.
 - g. Design bearings for minimum L_{10} life of 5 years in accordance with ABMA 9 or 11.
 - h. Provide neoprene seals with the bearings to prevent loss of lubrication and entrance of moisture and dirt.
 - 1) Labyrinth seals not acceptable.
- 7. Baseplate:
 - a. Designed to accommodate the anchorage design according to Drawing 40-D-6006.
 - b. Provide pumping unit with a structural steel baseplate supplied by the pump manufacturer.
 - c. Mount pump and motor on a common baseplate with separate mounting plates attached to baseplate for the pump and the motor.
 - d. Provide with suitable lifting lugs properly located to maintain a balanced load when handling the pump.
 - e. Sufficient size and rigidity to maintain the pump and motor in proper alignment and position without the benefit of epoxy grout, when subjected to the stresses imposed by normal operation, when pump dead headed, or during reverse flow following emergency shutdown.
 - f. Machine contact surfaces between the pump and mounting plate on the baseplate and between the motor and mounting plate on the baseplate to within 0.002-inch of true flat, level to 0.002-inch per linear foot.
 - 1) No welding after final machining.

- g. Transmit the loads from the baseplate to the concrete structure.
- h. Provide with standard NPT threaded drain connections and grout holes.
- i. Shaft offset and angularity between the pump and motor not to exceed coupling manufacturers' recommendations.

2.04 MOTOR

- A. Type: Horizontal, induction squirrel-cage designed for duties specified.
- B. Nameplate Ratings:
 - 1. Voltage: 480, 3-phase, 60-hertz.
 - 2. Duty: Continuous.
 - a. Capable of continuous operation under full load across the VFDs operational speed range.
 - 3. Service factor: 1.15 minimum.
 - 4. RPM: Motor speed shall meet pump speed requirement.
- C. Service Conditions:
 - 1. See Section 1.03, B, 6.
- D. Enclosure: Totally-enclosed, fan cooled (TEFC).
- E. Horsepower Rating: Sufficient to carry continuously the maximum possible pump load developed under specified conditions without benefit of service factor. Horsepower rating to meet Table 33 12 73A listed application values.
- F. Temperature Rise:
 - 1. Armature winding: Not to exceed 105 degrees Celsius when measured by resistance.
 - 2. Other motor parts: In accordance with NEMA MG 1.
- G. Starting Method: The motor shall be started and controlled via a Variable Frequency Drive (VFD).
- H. Starting Capability:
 - 1. Number of starts: Two starts in succession (coasting to rest between starts) with motor initially at ambient temperature or one start with motor initially at a temperature not exceeding its rated load operating temperature.
 - 2. Estimated maximum number of starts per day: 8.
- I. Nameplate Marking: As listed in NEMA MG 1 for alternating-current polyphase squirrel-cage motors.

- J. Stator:
 - 1. Stranded copper conductors.
 - 2. Insulation: Class F.
 - 3. Coils completely sealed and moisture-tight.
 - 4. Winding configuration: Manufacturer's standard wye.
 - 5. Completed winding subjected to two vacuum pressure impregnations in a solventless epoxy resin.
- K. Rotor: Squirrel-cage windings of suitable impedance for starting under specified conditions.
- L. Bearings:
 - 1. Motor shall be equipped with premium-quality deep groove ball bearings suitable for continuous-duty operation.
 - 2. To mitigate VFD-induced bearing currents:
 - a. The non-drive end bearing shall be electronically insulated (ceramic-coated or hybrid ceramic).
 - b. The drive end shall include a shaft grounding ring.
 - 3. Bearing shall be greaseable with Zerk fittings.
- M. Indicating and Protective Devices:
 - 1. Resistance Temperature Detector (RTD): 100 ohm, platinum, 3-wire.
 - 2. Location:
 - a. Two (2) RTDs per phase, located in stator slots between coil sides.
 - b. One (1) RTD for each motor bearing.
 - 3. Provide oil sight glass for oil lubricated bearings.
- N. Main Lead Terminal Box:
 - 1. Size: Oversized and suitable for terminating main leads with stress cones.
 - 2. Provide clamp-type ground lug suitable for connecting conductor shields inside terminal box.
 - 3. Manufacturer's standard cover with gasket.
 - 4. NEMA 4X rated.
- O. Low-Voltage Terminal Box:
 - 1. Size: Suitable for terminating leads from indicating and protective devices, and motor accessories.

2. Provide terminal blocks for terminating indicating and protective devices, and motor accessories.
3. Manufacturer's standard cover with gasket.
4. Provide with back panel for mounting terminal blocks.
5. NEMA 4X rated.

P. Accessories:

1. Grounding Provisions: Provide pad with drilled and tapped bolt holes on motor enclosure. Pad to be machined flat and left unpainted.
2. Provide means for lifting the entire pumping unit and/or individual components. This will entail below-the-hook devices required to lift these components. Lifting eyes attached to motor housing that are suitable for attaching slings for lifting with a crane. Lifting hooks or similar devices are not acceptable.
3. Wire Markers:
 - a. Type: Machine-printable, self-laminating label or tubular heat-shrink sleeve label.
 - b. Color: White.

2.05 VARIABLE FREQUENCY DRIVE (VFD)

A. Ratings:

B. VFD model or equal: Galt Electric G540-03050UL-03 Series Normal Duty.

1. Derate for operating elevation listed in Section 1.03, B, 6.
 - a. Rated for operating 100 HP Motor.
 - b. Voltage: 480V, 3-phase.
2. Provide suitable NEMA-rated control cabinet.

C. Standards:

1. Conform to IEC 61800, IEEE 519, NEMA IS 10033 and NFPA 70.
2. UL 508 Listed.

D. Functionality:

1. Start-up assistant.
2. Variable speed control with adjustable acceleration and deceleration ramps with up to seven preset speeds.
3. Automatic reset after power loss or fault.
4. Motor parameter tuning (auto-tune or manual entry).
5. PID Sleep function.

- E. Operator Control Panel:
 - 1. Graphical Display, keypad with keys for Hand, Off, and Auto.
 - 2. Start/Stop Control.
 - 3. Speed reference adjustment (via keypad).
 - 4. Real-time display of:
 - a. Output Frequency (Hz).
 - b. Output voltage and current.
 - c. Motor status (running, stopped, faulted).
 - d. Fault and alarm codes.
 - 5. Navigation buttons for parameter access and configuration.
 - 6. Password protection for parameter changes.
- F. Motor Protection Functions:
 - 1. Overload protection (I^2t -based).
 - 2. Overtemperature protection via:
 - a. Motor RTD inputs.
 - 3. Phase loss and imbalance detection.
 - 4. Ground fault protection.
 - 5. Short circuit protection.
 - 6. Overvoltage and undervoltage protection.
 - 7. Current limit and torque limit functions.
 - 8. Stall protection.
- G. Input/Output and Integration:
 - 1. 6 Digital Inputs 12 to 24VDC.
 - 2. 3 form C relay contact digital outputs.
 - 3. 2 Analog inputs, selectable from 0-10V or 0/4-20mA.
 - 4. 2 Analog Outputs 0/4-20mA.

2.06 FUSED DISCONNECT SWITCH

- A. Rated for 100 HP Motors:
 - 1. Max current: 300A.
 - 2. Fuse size and type: 225A Dual Element
- B. Requirements:

1. Flange-mounted fused, 3-pole, single-throw.
2. 480V.
3. Type HD (Heavy Duty).
4. Conform to NEMA KS 1.
5. Enclosure Type NEMA 4.
6. Lockable in open position.
7. Clearly labeled with motor and VFD identification.

2.07 EMERGENCY STOP (E-STOP) SWITCH

- A. Mushroom-head, maintained-contact type (push-to-stop, twist-to-release).
- B. Wired to the VFDs digital input configured for Safe Torque Off (STO) of Fault trip.
- C. Enclosure type NEMA 4.
- D. Labeled “Emergency Stop – Pump Motor”.

2.08 BREAKER AND CONDUCTOR RECONFIGURATION

- A. 100 HP VFD and Motor:
 1. Minimum Conductor Size Upstream of VFD: 2/0 AWG XHHW-2.
 2. Minimum Conductor Size Downstream of VFD: 2/0 AWG XHHW-2, Shielded.
 3. Minimum Ground Conductor Size: 6 AWG XHHW-2, insulation color or marked ground in accordance with NFPA 70.
 4. Minimum Conduit Size: 2-1/2 inch Rigid Metal Conduit (RMC).
- B. Emergency Stop:
 1. Minimum Conductor Size: 14 AWG, shielded.
 2. Minimum Conduit Size: 1/2-inch Rigid Metal Conduit (RMC).

2.09 CONTRACTOR SOURCE QUALITY TESTING

- A. Motor Shop Test:
 1. To be performed prior to pump shop assembly and testing.
 2. Assemble and test motor at the manufacturer’s shop to establish that performance requirements of these specifications and warranties under this contract have been fulfilled.
 3. Perform manufacturer’s routine factory tests and the following routine tests required by Part 20 of NEMA MG 1.
 - a. Measurement of winding resistance.

- b. No-load motoring readings of current, power, and nominal speed at rated voltage and frequency.
 - c. Alternating-current (AC) high-potential test. A direct-current (DC) high-potential test is not acceptable.
 4. Perform tests in accordance with requirements of NEMA MG1 and IEEE 112.
- B. Pumping Unit Shop Assembly and Test Requirements:
 1. Assemble mating parts of pumping unit (pump and motor). Match-mark pump components before disassembly for shipment, unless shipped assembled, to ensure correct field assembly.
 2. Prepare and submit certified pump performance curves showing overall efficiency, pump efficiency, horsepower input to the pump, and pump discharge at heads from the minimum head, specified in Table 33 12 73A - Performance Requirements, to pump shutoff head. Present test data at full-speed rating.
 - a. Pump certified performance curves to meet HI 14.6 and a 1U acceptance grade.
 3. Perform test with job pump and motor.
 4. Government to witness pump test. Give a 30-day notice to the Government before the pump and motor are ready to be tested.
 5. Certified hydrostatic test reports, per HI 14.6 and as modified herein:
 - a. Parts to be pressure tested at a minimum of 1.5 times the shutoff head pressure.
 6. Perform baseplate deflection analyses in accordance with HI 9.6.8.
 7. Perform thermal growth analyses of the pump and motor shafts.

2.10 WITNESS INSPECTION POINT (HOLD POINT)

- A. Provide 30-day notice before the start of the witness point milestone for Government technical engineers or Government representative to verify:
 1. Hydrostatic pressure testing.
 2. Motor shop test.
 3. Pumping unit shop assembly and test.
- B. Inspections to be conducted at a location that is approved for travel by the Department of State (www.travel.state.gov), without additional protective services or safety requirements.
 1. Provide necessary equipment and tools to perform an adequate inspection of the runner and associated equipment.
 2. If the inspection location does not meet these requirements, relocate the components/equipment within the United States, for inspection.

3. This relocation will be at the expense of the Contractor.

2.11 ACCESSORIES

A. Lubricating Grease:

1. In accordance with pump and motor manufacturer's recommendations.

PART 3 EXECUTION

3.01 PUMP AND MOTOR MANUFACTURER'S ERECTION ENGINEER

- A. No work is to be performed on pump or motor without applicable Manufacturer's Erection Engineer being on-site to direct/oversee the work.
- B. Manufacturer's qualified erection engineer to provide technical assistance to the Contractor during installation of the equipment. Applicable Manufacturer's Erection Engineer is required to be on-site, at minimum, for the following operations, for each pumping unit:
 1. Setting of pump and motor to baseplate.
 2. Installation of the pump.
 3. Installation of motor.
 4. Installation of baseplate, anchorage, and grouting.
 5. Coupling and alignment of the pumping unit.
 6. Functional and operational testing.

3.02 FIELD WITNESS INSPECTION POINT (HOLD POINT)

- A. Provide 7-day notice before the start of the witness point milestone for Government technical engineers or Government representative to verify:
 1. Installation of baseplate and anchorage prior to grouting, verifying pump alignment with piping.
 2. Setting of the pump and motor to the baseplate.
 3. Alignment of the pumping unit.
 4. Functional and operational testing.

3.03 EPOXY GROUT INSTALLATION

- A. Mix and install grout in accordance with manufacturer's instructions to fill space to be grouted:
 1. Concrete shall be a minimum of 28 days old.
 2. Clean and roughen concrete surfaces.

3. Remove dirt, dust, oil, grease, debris, paint, curing compounds, sealers, and unsound concrete.
4. Mechanically prepare concrete surfaces in accordance with manufacturer's instructions to give surface profile of a minimum of 1/8 inch and expose coarse aggregate of concrete.
5. Concrete shall have an open surface texture.

B. Placement:

1. Place grout at air temperature recommended by manufacturer.
2. Bring materials to be epoxied as close to 70 degrees F as possible.
3. Do not place grout over frozen concrete.
4. Hold grout in place with forms. Prepare forms to prevent sticking of grout.
5. Minimum 3/4-inch by 3/4-inch, 45-degree chamfer.
6. Mix grout components in accordance with manufacturer's instructions and technical data sheet.
7. Place grout in accordance with manufacturer's instructions.
8. Finish surface of grout in accordance with manufacturer's instructions.
9. Smooth sharp edges.

C. Protection:

1. Protect placed grout from damage during construction.
2. Maintain grout at 50 degrees F or greater for 72 hours, minimum, after placement unless otherwise recommended by manufacturer:
 - a. After protection period, discontinue protection against cold weather such that the drop in temperature of the grout will be gradual and will not exceed 5 degrees F per hour and 40 degrees F in 24 hours.
3. The more stringent of grout manufacturer's recommendation or this section shall apply.
4. Initial Loading:
 - a. Do not apply loads to epoxy grout sooner than 72 hours after placement:
 - 1) Apply load after epoxy grout has attained a compressive strength of 3,000 psi, minimum.
 - b. Time required for epoxy grout used to attain this strength will be determined by COR.
 - c. Take care when applying loads on hardened epoxy grout.
 - d. Contractor shall be responsible for damage resulting from impact loads when positioning equipment or metalwork.

- D. Protect from movement until grout has fully cured.

3.04 EQUIPMENT INSTALLATION

- A. Service each pumping unit before installing.
1. Servicing consists of complete lubricating, adjusting, and cleaning of the unit.
 2. Lubricating grease furnished by the Contractor in accordance with the pump and the motor manufacturers' recommendations.
- B. Responsible for correct assembly and alignment of parts of each pumping unit.
- C. Install pumping units in accordance with pump and motor manufacture's recommendations except as modified herein.
1. Install Pumping Unit in accordance with Drawing 40-D-6006. Anchor into existing foundation.
- D. Apply a suitable lubricant to the pump baseplate leveling screws and pump baseplate anchor bolts to prevent bolt threads from seizing to the epoxy grout.
- E. After servicing, installation of the pumping unit to be done by a qualified millwright, under the supervision of the Manufacturer's Erection Engineer, and in accordance with HI 14.1 - 14.4 and manufacturer's installation instructions.
- F. Tighten baseplate anchor bolts, pump mounting bolts, and motor mounting bolts to manufacturer's recommended torque, prior to beginning the field tests on each unit. At the conclusion of the field tests on each unit, re-check the respective bolt torques. If the bolts are insufficiently torqued, the Government may direct the tests be conducted again, at no additional cost to the Government.
- G. Carefully handle equipment so that no part will be bent, broken, or otherwise damaged.
1. Repair damage caused by the Contractor, by and at the expense of the Contractor, as directed and approved by the Government.
 2. Do not install damaged or defective parts.
- H. Couplings and Flanges:
1. Thoroughly clean dirt and burrs from couplings, flange faces, and machined surfaces before connection to ensure tight fit and true alignment.
 2. Coat finished surfaces of flanged joints with joint compound before bolting.
 3. Tighten flange bolts several times at intervals until the initial stretch is taken up and it is positively ensured that there will be no leakage.
- I. Baseplate and Pump/Motor Installation:
1. Remove pump and motor from baseplate before beginning baseplate installation.

2. Thoroughly clean the baseplate before installation to remove paint, dirt, rust, and other foreign matter.
3. Level and set the elevation of the baseplate by means of leveling bolts to within 0.002-inch per linear foot under each working component of the unit (0.002-inch per linear foot under the pump and 0.002-inch per linear foot under the motor).
4. Install baseplate assembly so that mating surfaces of the pump feet and motor feet to the mounting plates have 100-percent contact. No shims, wedges, or lead washers allowed.
5. Set pump on mounting plate of baseplate and align the pump and connect to the suction and discharge piping. Check baseplate level again as required.
6. Final installation of the pump to be at the elevation and position that lines up the suction and discharge piping with the pump and shall account for the offset between the centerline of the suction and discharge connections on the pump and respective connections on the intake and discharge manifolds.
7. Set motor on mounting plate of baseplate and align pump and motor. Secure baseplate by tightening the upper nuts on the embedded anchor bolts evenly and using the leveling bolts.
 - a. Perform alignment using dial indicators or laser alignment to check both parallel and angular misalignments.
 - b. Maximum allowable total angular run-out, with indicating point approximately 12 inches apart, is 0.003-inch.
 - c. Maximum allowable total indicated parallel run-out is 0.004-inch.
8. Grout baseplate in place using approved epoxy grout.
9. Perform grout placement in a manner to ensure 100-percent contact between the baseplate and the concrete pad.
10. After epoxy grout has set, fully tighten upper nuts on the embedded anchor bolts and recheck alignment of the pump and motor.
 - a. Take corrective measures, as required, until alignment and leveling tolerances meet specification requirements.
11. Fully tighten nuts on embedded anchor bolts to manufacturer's recommended torque value.
12. Check level of baseplate and piping alignment after grouting and tightening nuts. Install the motor and align the motor shaft to the pump shaft.
13. After final alignment, dowel pump and motor to their respective mounting plates on the common baseplate.
14. Turn motor shaft and pump shaft by hand to ensure free rotation in their bearings.
15. After determining correct direction of rotation, connect water piping and make electrical connections.

16. Connect pump shaft and motor shaft and make necessary adjustments in accordance with manufacturer's instructions after electrical connections have been made and correct direction of rotation has been determined.
17. After installing completely, service each pumping unit.
 - a. Completely lubricate, adjust, and clean each unit.
 - b. Check lubricating devices for correct operation before thoroughly lubricating unit.
 - c. Lubricating grease to be furnished by the Contractor in accordance with the pump and motor manufacturers' recommendations.

J. Electrical Equipment:

1. Contractor shall install the new VFD, fused disconnect switches, and emergency stops devices in accordance with the following requirements.
2. Remove existing disconnect, control system, and cable to motor. If conduit meets minimum size in section 2.08, A, 4, it may be reused, otherwise remove and replace conduit with section size requirement.
3. New VFD shall be installed in same physical location as previously removed VFD.
 - a. Mounting shall be secure and compliant with manufacture recommendations.
 - b. All terminations shall be clearly labeled and torqued to specifications.
4. New Emergency Stops shall be installed near where motor is located and a second on upper platform.
 - a. E-Stops shall be wired to the VFDs configured to initiate motor stop per VFD manufacturer instructions.
5. Fused disconnect switch shall be installed in same location as the previous disconnect switch was installed.
6. Complete section 3.05L prior to making final connections to motor.
7. Contractor shall furnish and install all conduit and conductors meeting minimum specification in section 2.08A from:
 - a. Fused disconnect load side to the VFD.
 - b. On line side of fused disconnect utilizing UL approved stacked lugs ensure blower motor disconnect conductors are connected.
 - c. VFD to the motor.
 - d. E-Stop to the VFD control terminals.
 - e. Reuse existing conduit from VFD to upper control panel.
8. Terminate armature leads and external motor cables. Make terminations in accordance with manufacturer's instructions and recommendations.

9. Terminate leads for motor accessories and external control cables. Make terminations in accordance with manufacturer's instructions and recommendations.
10. All wiring shall be installed in accordance with NFPA 70 and local code requirements.
11. Install fuses meeting minimum size in section 2.06A, 2 into fused disconnect.
12. Configure new VFD to operate motor over complete range. RTD input and monitoring alarms for VFD shall also be configured. VFD and motor will be left in fully functional state.
13. Integrate new VFDs control panel for start/stop and speed control.
14. Contractor shall coordinate with Government personnel to ensure proper handoff and readiness for final connection.

3.05 SERVICING AND FIELD TESTING

- A. Coordinate operational field testing procedures with Government.
- B. Before performing operational field tests, completely lubricate, adjust, and clean the pumping unit.
- C. Furnish lubricating devices in accordance with the pump and motor manufacturers' recommendations.
 1. Check lubricating devices for correct operation before thoroughly lubricating unit.
- D. If necessary, furnish and install temporary instruments, gauges, recording and sensing devices, and special brackets required for the operational field tests.
- E. Clean up the testing area before and after each test.
- F. Schedule construction work so that these tests can be conducted.
- G. Have the necessary craftsman available to make necessary changes (either temporary or permanent).
- H. Responsible for operating the unit, recording the data, and enforcing safety procedures.
- I. Schedule, conduct, and coordinate tests required to be witnessed by the Government.
- J. Perform tests under the direction/oversight of the Manufacturer's Erection Engineer, and in accordance with HI 14.6 and manufacturer's operational field testing procedures.
- K. Clean internal surfaces of system components to ensure the effective removal of contaminants. Remove loose solid contaminants. Remove rust from all surfaces. Furnish materials, supplies, labor and equipment for cleaning the system components.

- L. Before making electrical connections and energizing the motor, perform the following checks and tests:
 - 1. Successfully test all motor control, monitoring, and protective circuits.
 - 2. Insulation resistance test in accordance with IEEE 43.
 - 3. AC high-potential test in accordance with NEMA MG 1 and IEEE 112.
 - a. A DC high-potential test is not acceptable.
 - b. If motor fails test, correct in accordance with manufacturer's recommendations.
- M. Perform VFD start-up per manufacturer's instructions.
 - 1. Perform stepped ramping of VFD in accordance with 3.05N verifying that VFD speed and motor operation with no overload.
 - 2. Perform stepped ramping of VFD from SCADA system to ensure operation and control.
- N. Perform an operating test on pumping unit under load to determine that the unit is correctly installed and meet the flow requirements.
 - 1. Supply all testing equipment.
 - 2. Test conducted by the Contractor and witnessed by the Government for pumping unit.
 - 3. During the test, observe pumping unit operation and record noise level, vibration analysis, shaft alignment and run-out measurements, pump and motor bearing temperatures, motor voltage and amperage, and motor stator temperature.
 - a. Vibration analysis to be full spectrum analysis using a monitor that records and prints analysis results.
 - 1) Vibration levels not to exceed 1/2 the acceptable field vibration limits established by HI 9.6.4.
 - 4. Without additional cost to the Government, Contractor shall make changes and correct errors for which the Contractor is responsible.

END OF SECTION